

SKUASK-K UET-2017
QUESTION BOOKLET

Booklet
Series

C

Booklet No.

106903

TIME: 2 HOURS 15 MINUTES FOR PCB & PCM and 3 HOURS FOR PCBM

This booklet contains: 20 Pages

Do not open this booklet until you are asked by the invigilator.

PCB=92

INSTRUCTIONS

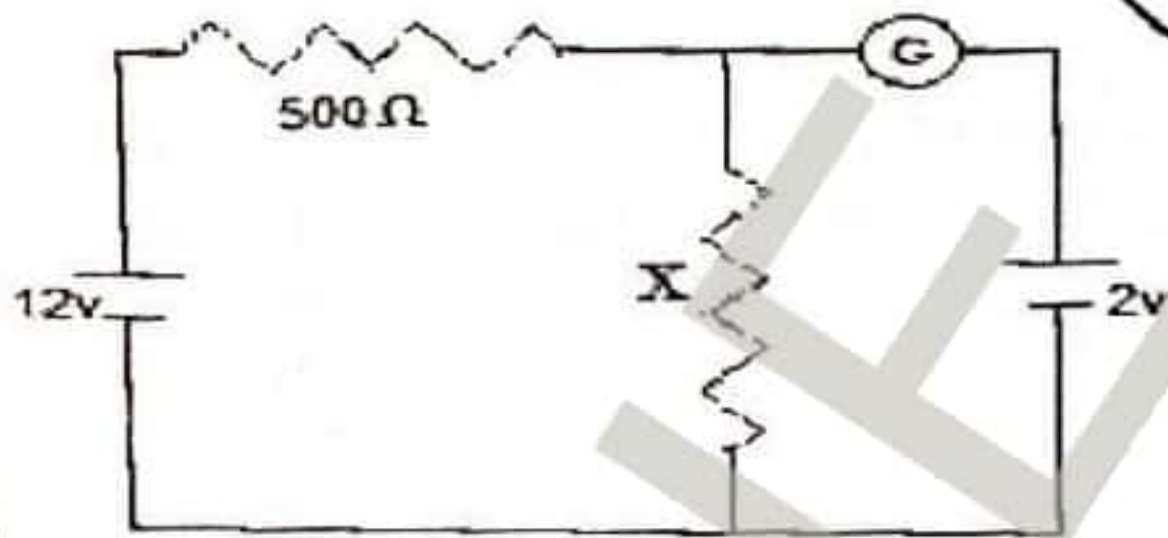
1. After opening the Question booklet, ensure that it contains all pages and questions. **In case of any discrepancy, get the question booklet replaced immediately.**
2. There are **180 MCQs** under Physics, Chemistry, Biology and Mathematics with **45** questions in each section. Students appearing under test stream **Biology (PCB)** will attempt questions numbered between **1-135**, those appearing under test stream **Math (PCM)** will attempt questions numbered between **1-90** and **136-180** and those appearing under test stream **Both (PCBM)** will attempt all the **180** questions.
3. Use only **blue/black** pen for writing your:
 - i. Particulars viz. **Name, Roll Number, Q. Booklet Series** and **Q. Booklet Number** on the OMR answer sheet in respective columns. OMR answer sheets without Q. Booklet Series and Number will not be evaluated.
 - ii. Responses on the OMR answer sheet.
4. **Do not write anything** else on OMR answer sheet. For rough work use the spaces provided on question paper booklet. Erasing, cutting or overwriting on OMR sheet will be deemed as incorrect answer and shall result in zero marking.
5. The candidates should on demand show their admit cards to invigilators or any other authorized representative of SKUAST-K.
6. No candidate will be allowed to leave the examination hall before half time except under special circumstances and permission by Center Superintendent.
7. The candidates shall hand over the OMR answer sheet to the invigilator before leaving the examination hall.
8. Use of *electronic* gadgets such as calculators, mobiles and pagers etc. is not allowed in the examination hall.
9. The cases pertaining to the use of unfair means will be governed by the rules and regulations of SKUAST-K.
10. Any act of **indiscipline/ misconduct/ use of unfair means** shall result in **forfeiture** of candidate's right to sit in the examination. The decision of Superintendent of the centre in this regard shall be final.

L2
1A

-2
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Physics (1-45)

1. In the given circuit galvanometer reads zero, the value of resistance "X" is:



- a) 10 Ω
b) 100 Ω
c) 500 Ω
d) 200 Ω

2. A charged particle moving in a uniform magnetic field penetrates a layer of lead thereby loses half of its K.E, the radius of circle changes as: (r is the initial radius of circle)

- a) $r' = r/\sqrt{2}$ b) $r' = r\sqrt{2}$ c) $r' = r$ d) $r' = r/2$

3. A wire of length "L" meters carrying a current "I" Amperes is bent in the form of the circle. The magnitude of magnetic moment of the wire is:

- a) $\frac{I^2 L}{2\pi}$ b) $\frac{I^2 L}{4\pi}$ c) $\frac{IL^2}{2\pi}$ d) $\frac{IL^2}{4\pi}$

4. A galvanometer of range 3v and resistance 200 Ω cannot be converted into an Ammeter of range:

- a) 10 mA b) 100 mA c) 1 A d) 10 A

5. An A.C circuit containing an inductor of inductance $L = 0.5$ H and capacitor of capacity $C = 8 \mu\text{f}$ in series. The current in the circuit is maximum when angular frequency is:

- a) 100 b) 200 c) 300 d) 400

6. A jet plane is travelling towards west at a speed of 1800 km/hr. What is the potential difference between the ends of the wing having a span of 25 m, if the Earth's magnetic field at the location has a magnitude of 5×10^{-4} T and the angle of dip is 30° :

- a) 1.75v b) 3.12v c) 6.25v d) 8.5v

7. In Yong's double slit experiment, using red and blue lights of wave lengths 600 nm and 480 nm respectively, the value of "n" for which the nth red fringe coinciding with (n+1)th blue fringe is:

- a) 5 b) 4 c) 3 d) 2

8. Two coherent monochromatic light beams of intensities "I" and "4I" are superimposed. The maximum and minimum possible intensities in the resulting beam is:

- a) 5I, I b) 5I, 3I c) 9I, I d) 9I, 3I

$$5 \times 10^{-4} \frac{1}{2}$$

$$\beta = \frac{\lambda D}{d} \quad y = \frac{\lambda d}{D}$$

$$\frac{600}{480} = \frac{(n+1)}{n} (\sqrt{I} + \sqrt{4I})^2$$

$$1.75 - 2.57 \quad \dots \quad 2.5 \times 10^{-2} \quad (2.25)$$

9. A fish rising vertically with a speed of 3 m/s to the surface of water sees a bird diving vertically towards it with a speed of 9 ms⁻¹. The refractive index of water $\mu = 4/3$, the actual velocity of dive is:

- a) 9 ms⁻¹ b) 6 ms⁻¹ c) 4.5 ms⁻¹ d) 12 ms⁻¹

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin 90^\circ}{\sin \theta}$$

10. Angle of prism is "A" and its one surface is silvered. Light rays falling at an angle of incidence 2A on the first surface, return back through the same path after suffering reflection at the silvered surface, the refractive index of the prism is:

- a) $2 \sin A$ b) $2 \cos A$ c) $\cos (A/2)$ d) $\tan A$

$$\sin i = \mu \sin r$$

$$\sin 2A = \mu \sin A$$

$$2 \sin A \cos A = \mu \sin A$$

11. Light of two different frequencies whose photons have energies 1 eV and 2.5 eV respectively, successively illuminate a metal of work function 0.5 eV. The ratio of maximum speeds of the emitted electrons will be:

- a) 1:5 b) 1:4 c) 1:2 d) 1:1

12. The orbital velocity of the electrons in the ground state of hydrogen atom is "V". The velocity when the electron is excited to the energy state of energy -1.51 eV is:

- a) $V/9$ b) $V/4$ c) $V/3$ d) $V/2$

13. Atomic mass number of an element is 232 and its atomic number is 90. The end product of this radioactive element is an isotope of lead (A = 208 and Z = 80). The number of α and β particles emitted is:

- a) (3,3) b) (6,4) c) (6,0) d) (1,6)

$$A_{90}^{232}$$

$$A = 232 - 4 \times 6 = 160$$

$$0.2 \times 0.6 \times 2 = 0.24$$

14. A p-type semiconductor has acceptor level 66 meV above the valance band. The maximum wave length of light required to create a hole is nearly equal to:

- a) 2×10^{-5} m b) 2×10^{-6} m c) 2×10^{-8} m d) 2×10^{-10} m

15. What is the voltage gain in a common emitter amplifier, where input resistance is 3 Ω and load resistance 24 Ω , $\beta = 0.6$?

- a) 8.4 b) 4.8 c) 2.4 d) 480

$$V = I \times R$$

$$0.6 \times \frac{24}{3} = 4.8$$

$$4.8$$

16. The dimensional formula for electromotive force is same as that of:

- a) Force b) Energy c) Potential d) Current

$$\text{Current} \times \text{Electricity}$$

17. A force "F" is applied on a square plate of side "L". If percentage error in the measurement of "L" is 2% and that in "F" is 4%, the percentage possible error in pressure is:

- a) 2% b) 4% c) 6% d) 8%

$$P = \frac{F}{A}$$

$$\frac{\Delta P}{P} = \frac{\Delta F}{F} + \frac{\Delta A}{A}$$

18. A boat which has a speed of 5 km/hour in still water crosses a river of width 1 km along the shortest path in 15 minutes. The velocity of river in km/hour is:

- a) 1 b) 3 c) 4 d) $\sqrt{41}$

$$V_r = \sqrt{5^2 - 4^2} = 3$$

$$V = \frac{d}{t} = \frac{1}{0.25} = 4 \text{ km/h}$$

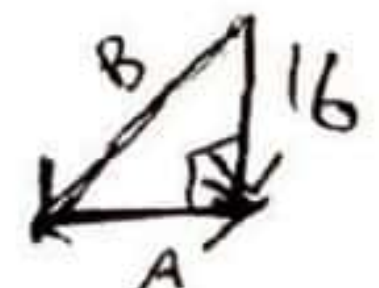
19. The sum of two forces acting at a point is 16N. If the resultant force is 8N and the direction is perpendicular to minimum force, then the forces are:

- a) 6N and 10N b) 8N and 8N c) 4N and 12N d) 2N and 14N

$$A + B = 16$$

20. At the top of the trajectory of a projectile, the direction of its velocity and acceleration are:

- a) Perpendicular to each other b) Parallel to each other c) inclined to each other at 45° d) Anti parallel to each other



$$16^2 - 8^2 = x^2$$

$$256 - 64 = x^2$$

$$15 = \frac{1 \text{ km}}{t}$$

$$A \cdot B$$

21. A block of metal weighing 2kg is resting on a frictionless plane. It is struck by a jet releasing water at a rate of 1kg/s and at speed of 5m/s. The initial acceleration of the block is:

- a) 2.5 m/s^2 b) 5 m/s^2 c) 10 m/s^2 d) 15 m/s^2

22. Essential characteristic for equilibrium is:

- a) Momentum equal to zero b) acceleration equal to zero
c) Kinetic energy equal to zero d) Velocity equal to zero

23. A lift is descending at a constant speed "V". A passenger in the lift drops a coin. The acceleration of the coin towards the floor will be:

- a) zero b) g c) -g d) V + g

24. If the force on a rocket moving with velocity of 300 m/s is 210N, then the rate of combustion of the fuel is:

- a) 0.7 kg/s b) 1.4 Kg/s c) 0.07 kg/s d) 10.7 kg/s

25. A stationary particle explodes into two parts of masses " m_1 " and " m_2 " which move in opposite direction with velocities " V_1 " and " V_2 ". The ratio of Kinetic energies " E_1 " and " E_2 " is:

- a) m_1/m_2 b) 1 c) $m_1 V_2/m_2 V_1$ d) m_2/m_1

26. A ball is dropped from a height of 20 cm. Ball rebounds to a height of 10 cm. The loss of energy is:

- a) 25% b) 75% c) 50% d) 100%

27. If a sphere is rolling, the ratio of its rotational energy to total kinetic energy is:

- a) 7:10 b) 2:5 c) 10:7 d) 2:7

28. A rubber ball filled with water is having a small hole. This is used as a bob of a simple pendulum. Then the period of such a pendulum:

- a) increases b) decreases
c) first increases then decreases d) first decreases then increases

29. A circular turn table has a block of ice placed at its centre. The system rotates with an angular speed " ω " about an axis passing through the centre of the table. If the ice melts on its own without any evaporation, the speed of rotation of the system:

- a) becomes zero
b) remains constant at same value of " ω "
c) increases to the value greater than " ω "
d) decreases to the value less than " ω "

30. If the temperature difference on the two sides of a wall increases from 10°C to 20°C , its thermal conductivity:

- a) remains unchanged b) is doubled
c) is halved d) becomes four times

31. The amount of the thermal radiation emitted from unit area of a perfectly black body in one sec say "E" is related to the absolute temperature "T" of the body as:

- a) $E \propto T$ b) $E \propto T^2$ c) $E \propto T^3$ d) $E \propto T^4$

32. A thin circular copper plate, a sphere and a cube of same mass and material are heated to 100°C temperature. If they are allowed to cool, which of the three will cool first:

- a) cube ☒ b) circular plate c) sphere d) all at same time

33. Water is used to cool the radiators of engines in cars because:

- a) of its low boiling point ☒ b) of its high specific heat
c) of its low density d) of its easy availability

34. Isothermal bulk modulus of a gas at atmospheric pressure is:

- a) $2.02 \times 10^5 \text{ N/m}^2$ ☒ b) $1.013 \times 10^5 \text{ N/m}^2$ c) 1 mm of Hg d) 13.6 mm of Hg

35. An ideal heat engine exhausting heat at 77°C is to have 30% efficiency. It must take heat at:

- a) 127°C ☒ b) 227°C c) 327°C d) 673°C

36. The total energy of simple harmonic oscillator is "E". The K.E of the oscillator when the displacement is half of the amplitude is:

- a) $\frac{\sqrt{3}E}{2}$ b) $E/2$ ☒ c) $E/4$ d) $E/3$

37. Two bodies "M" and "N" of equal masses are suspended from two separate springs of constants " K_1 " and " K_2 " respectively. If they oscillate with equal velocities, then the amplitudes of "M" and "N" are in the ratio of:

- a) K_1/K_2 b) K_2/K_1 ☒ c) $\sqrt{\frac{K_1}{K_2}}$ d) $\sqrt{\frac{K_2}{K_1}}$

38. Two waves represented by $y_1 = a \sin(\omega t + \pi/6)$ and $y_2 = a \cos(\omega t)$. If these waves superimpose, the amplitude of the resultant wave is:

- a) $\sqrt{2}a$ b) a ☒ c) $\sqrt{3}a$ d) $2a$

39. A tuning fork when sounded along with a standard source of frequency 300 Hz produces 5 beats/sec. The tuning fork when loaded by some wax is again found to give 5 beats/sec with the same standard source. The frequency of fork originally is:

- a) 290 Hz ☒ b) 295 Hz c) 305 Hz d) 310 Hz

40. Two charged metal spheres of capacitances " C_1 " and " C_2 " are brought in contact and then separated. The final charges " q_1 " and " q_2 " on them will satisfy the condition:

- ☒ a) $q_1 C_2 = q_2 C_1$ b) $q_1 C_1 = q_2 C_2$
c) $q_1 C_2 > q_2 C_1$ d) $q_1 C_2 < q_2 C_1$

41. A charge "q" is distributed over two spheres of radii "R" and "r" such that their surface charge densities are equal. The ratio of the charges on the spheres is:

- a) r/R ☒ b) r^2/R^2 c) r^3/R^3 d) r^4/R^4

42. If a charged spherical conductor of radius 10 cm has potential "V" at a point distance 5 cm from its centre. The potential at a point distant 15 cm from the centre will be:

- a) $V/3$ ☒ b) $2V/3$ c) $3V/2$ d) $3V$

43. An alpha particle is accelerated through a potential difference of 10^6 v, its K.E will be:

- a) 1 Mev ~~b) 2 Mev~~ c) 4 Mev d) 8 Mev

44. A 4μf capacitor is charged to 400v and it is connected to 1000 Ω resistor. The heat produced is:

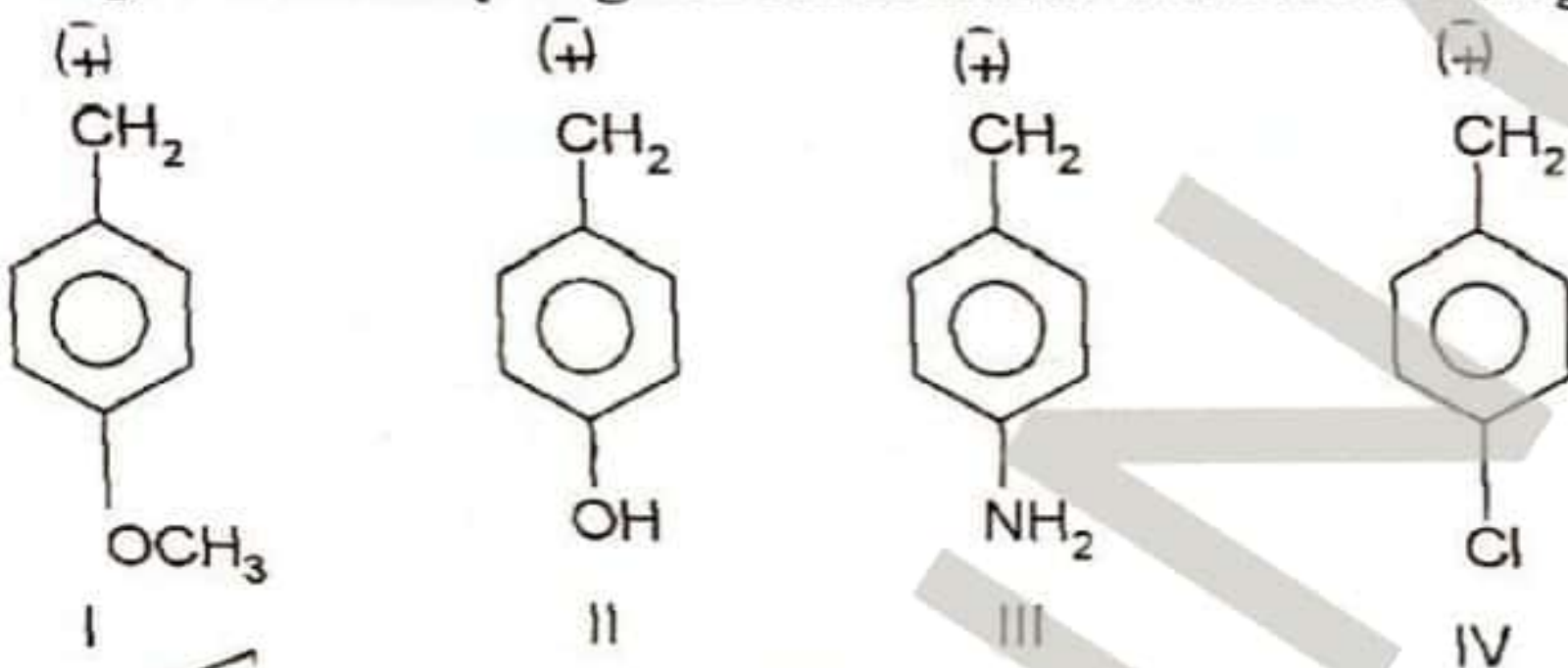
- a) 0.64 j ~~b) 0.32 j~~ c) 0.16 j d) 1.26 j

45. A storage battery of emf 8v and internal resistance 0.5 Ω is being charged by a 120v d.c supply using a series resistance of 15.5 Ω. The terminal voltage of the battery during charging is:

- a) 120v b) 8v ~~c) 11.5v~~ d) 120v

Chemistry (45-90)

46. Arrange the stability of given carbocation in the decreasing order:



- a) $I > II > III > IV$ b) $III > II > I > IV$
c) $IV > I > II > III$ d) $II > III > I > IV$

47. In the reaction sequence



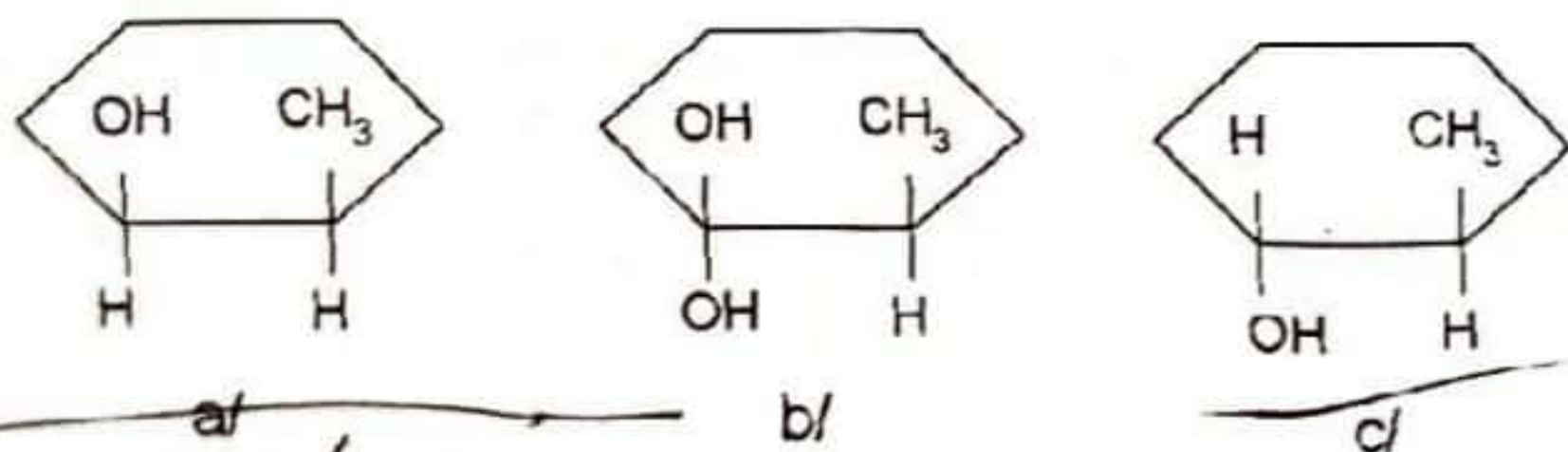
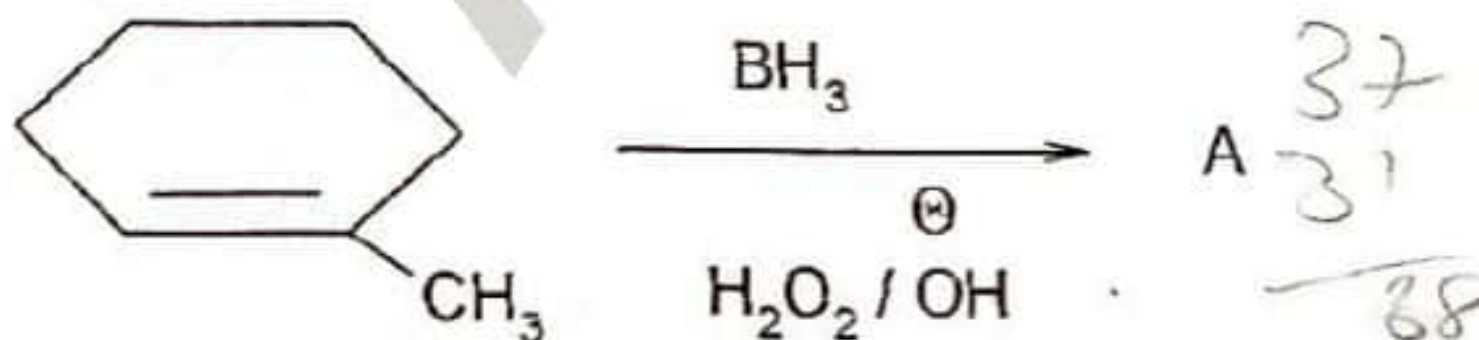
A and B are respectively:

- a) $CH_3CH=NH$ and CH_3CHO b) CH_3COOH and $CH_3CH_2NH_2$
c) CH_3CONH_2 and CH_3COOH d) $CH_3CH_2NH_2$ and CH_3CH_2OH

48. Which of the following not reduce Fehling solution:

- a) HCHO b) CH_3CHO ~~c) HCOOH~~ d) C_6H_5CHO

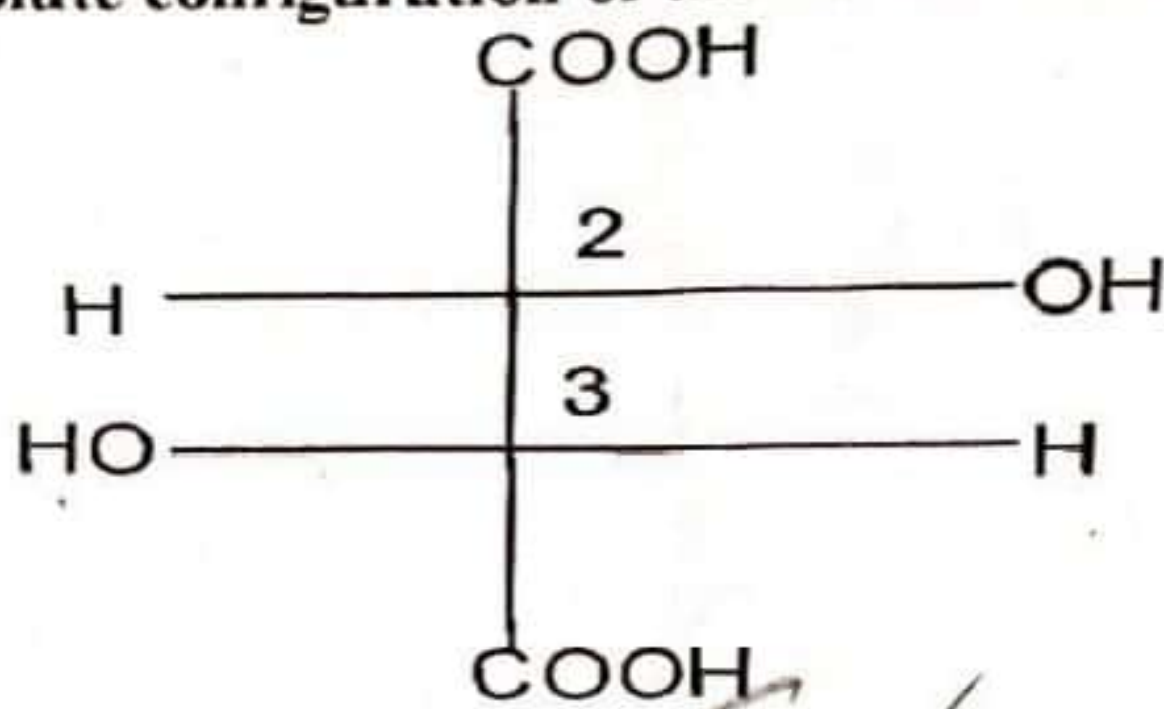
Identify the compound A in the following reaction



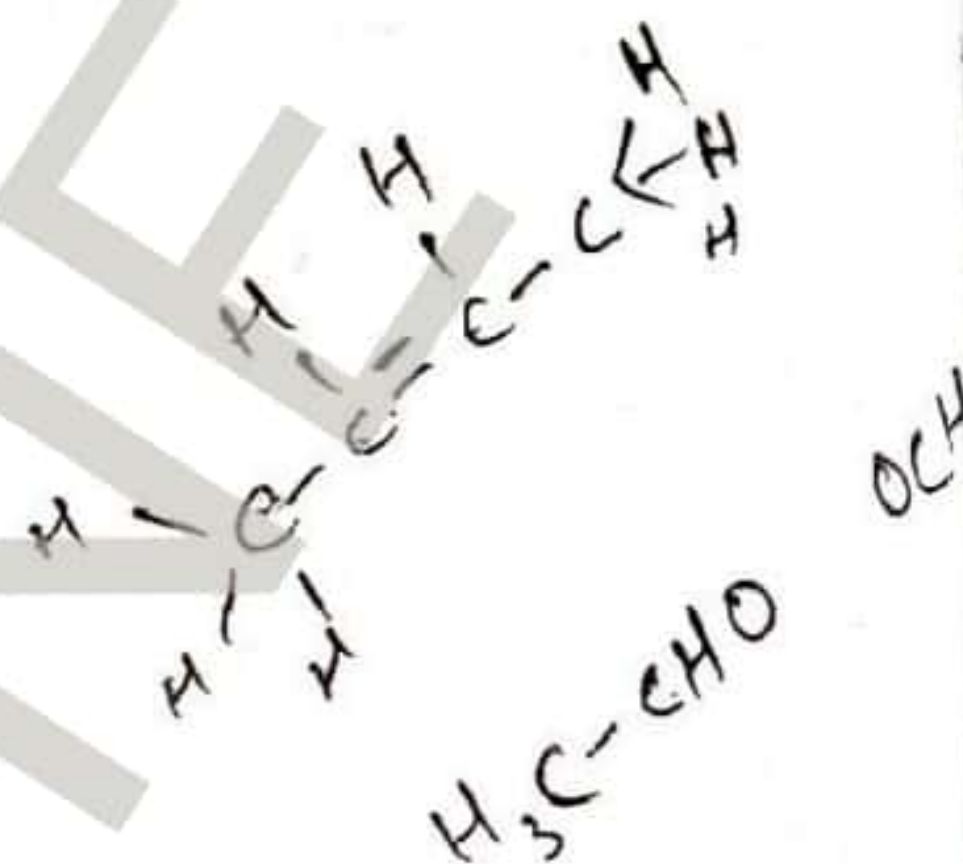
50. Which one of the following is optically active and gives yellow colour with Ninhydrin:

- a) ~~Proline~~ b) Glycine c) Alanine d) Phenyl alanine

51. The absolute configuration of the C₂ and C₃ atoms in the molecule



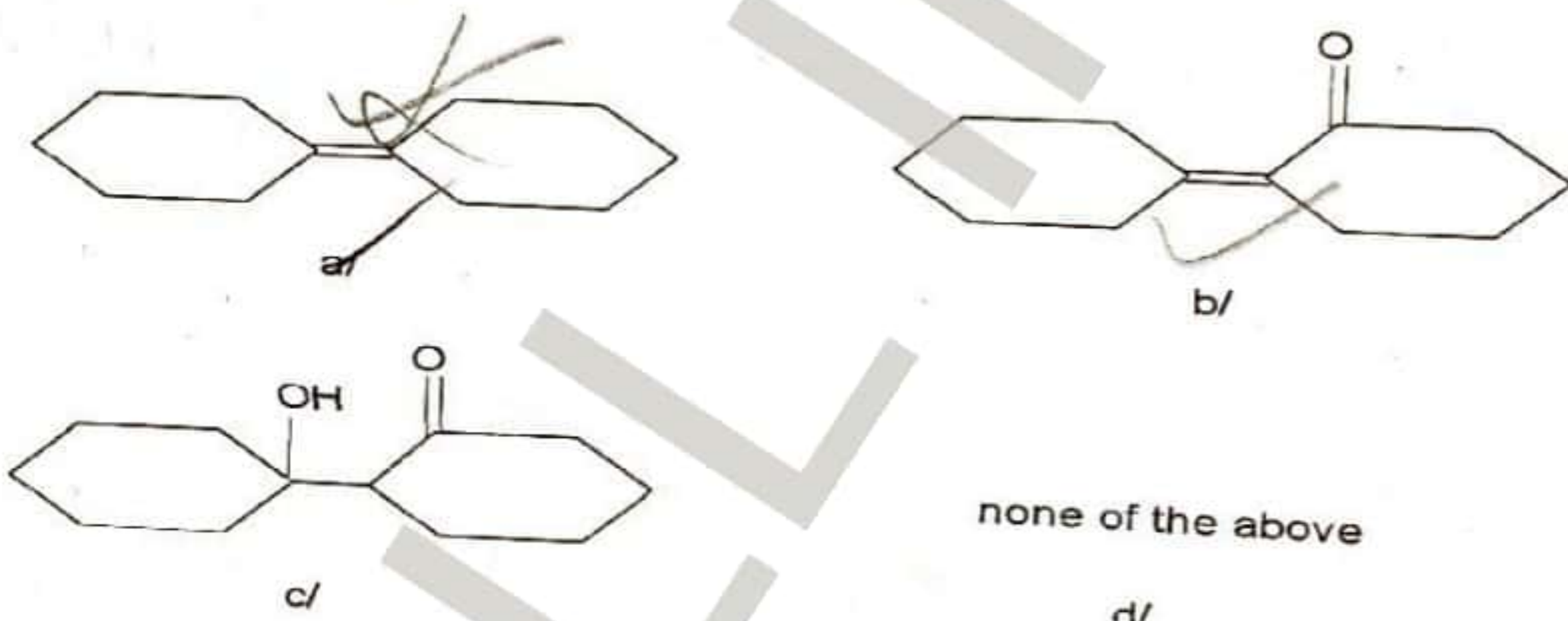
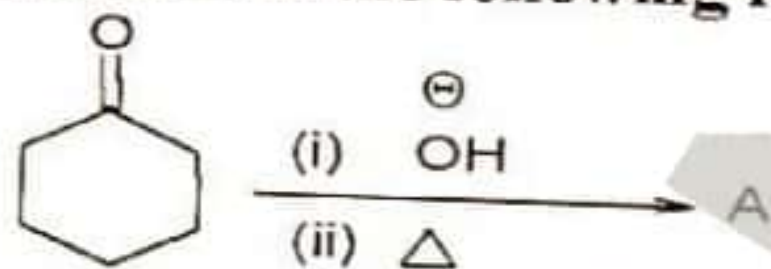
- a) S, S b) R, R c) R, S d) S, R



52. But-2-ene on ozonolysis gives:

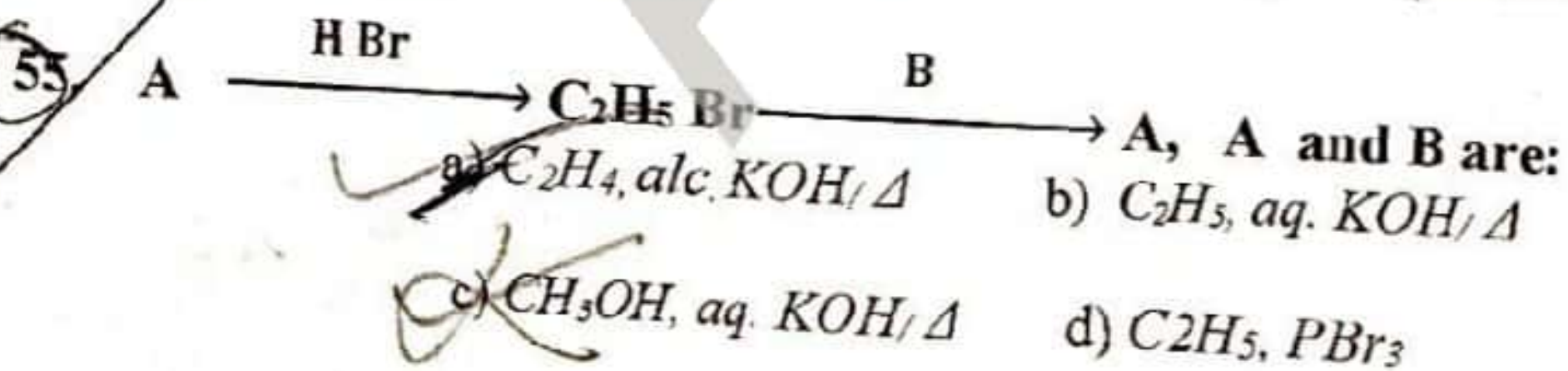
- a) Ehtanal only b) ~~mehtanal and Ethanal~~ c) Methanal only d) Propanal and butanol

53. Identify the product A in the following reaction:

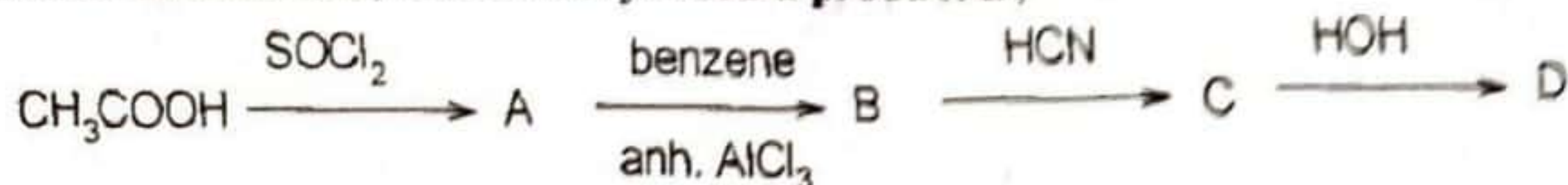


54. The most acidic among the following is:

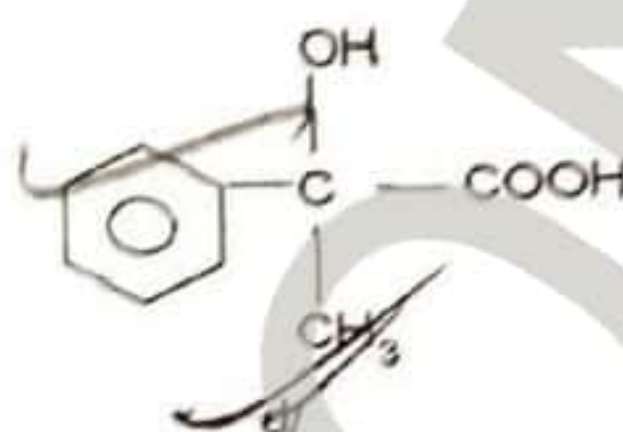
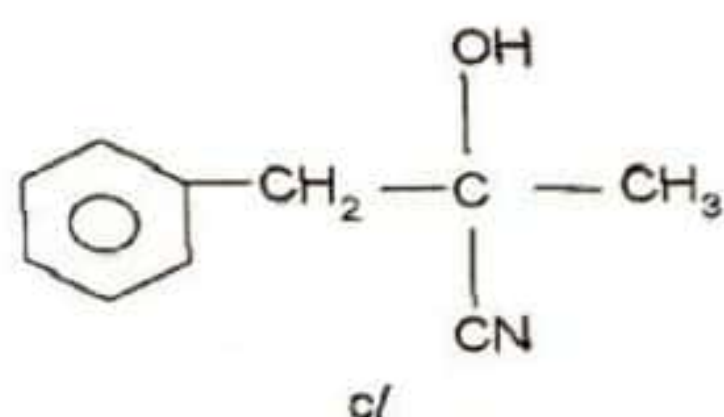
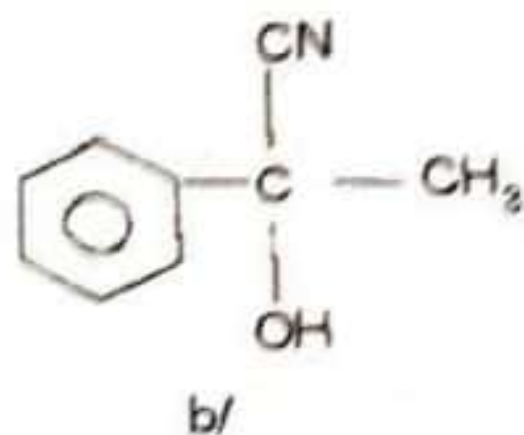
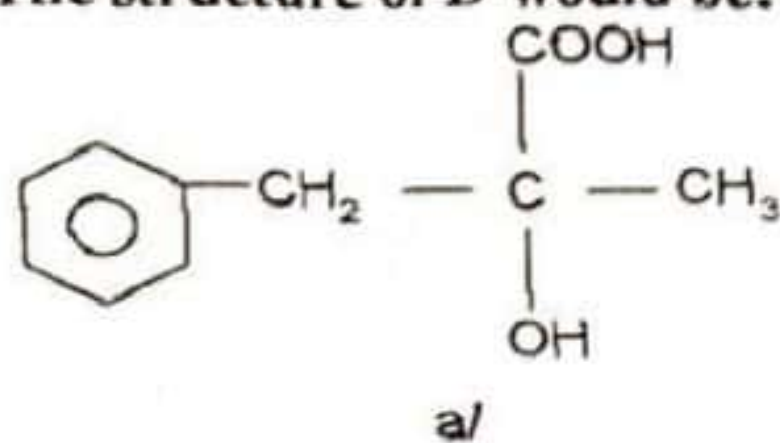
- a) p-Cresol b) O-Cresol c) p-nitrophenol d) p-chlorophenol



56. In a set of reactions, acetic acid yielded a product D,



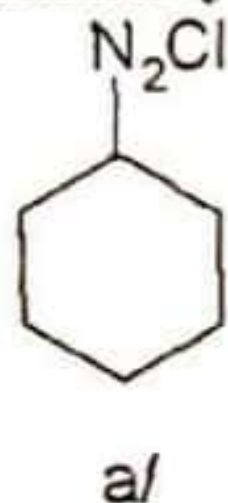
The structure of D would be:



57. Which is the correct order of decreasing strength?

- a) $\text{ClCH}_2\text{COOH} > \text{CH}_3\text{COOH} > \text{HCOOH} > \text{C}_6\text{H}_5\text{OH}$
 b) $\text{ClCH}_2\text{COOH} > \text{HCOOH} > \text{CH}_3\text{COOH} > \text{C}_6\text{H}_5\text{OH}$
 c) $\text{ClCH}_2\text{COOH} > \text{HCOOH} > \text{C}_6\text{H}_5\text{OH} > \text{CH}_3\text{COOH}$
 d) $\text{ClCH}_2\text{COOH} > \text{CH}_3\text{COOH} > \text{C}_6\text{H}_5\text{OH} > \text{HCOOH}$

58. Action of NaNO_2^+ dil HCl on ArNH_2 yields $\text{ArN}_2^+\text{Cl}^-$. A similar reaction with cyclohexyl amine will yield:



59. Propylamine yields a volatile compound X on warming with alc. Alkali and chloroform. X has an offensive odour. The structure of X is:

- a) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CN}$ b) $(\text{CH}_3)_2\text{CHCN}$
 c) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NC}$ d) $(\text{CH}_3)_2\text{CHNC}$

60. Mutarotation does not occur in:

- a) Sucrose b) D-glucose c) L-glucose d) none of these

61. Which of the following are possible values of n , l and m for an atom having maximum value of $m = +2$:

- a) $n = 4, l = 3, m = +2$ b) $n = 3, l = 2, m = -2$
 c) $n = 3, l = 3, m = +2$ d) $n = 2, l = 3, m = +2$

62. The orbital angular momentum of electron in p-orbital is:

- a) $\frac{n}{\sqrt{2}\pi}$ b) $\frac{h\pi}{\sqrt{2}}$
 c) $\frac{1}{\sqrt{2}} \times \frac{h}{2\pi}$ d) $\sqrt{3} \times \frac{h}{2\pi}$

$$\sqrt{l(l+1)} \frac{h}{2\pi}$$

$$\sqrt{1(1+1)} \frac{h}{2\pi}$$

$$\sqrt{2} \frac{h}{2\pi} = \frac{\sqrt{2}}{2} \frac{h}{2\pi}$$

$$\frac{h}{\sqrt{2} \cdot 2\pi}$$

63. The pH of 10^{-8} molar solution of HCl in water is:

- a) between 4 and 6
c) 8

- b) between 6 and 7
d) between -6 and -8

64. The equilibrium constant value K_p for the equilibrium $H_{2(g)} + I_{2(g)} \rightleftharpoons 2HI_g$ changes with:

- a) total pressure
c) catalyst

- b) temperature
d) the amount of H_2 and I_2 present

65. Which of the following is not a buffer solution:

- a) $0.8 M H_2S + 0.8 M KHS$
c) $3 M H_2CO_3 + 3 M KHCO_3$

- b) $2 M C_6H_5NH_2 + 2 M C_6H_5NH_3^+ Br^-$
d) $0.05 M KClO_4 + 0.05 M HClO_4$

66. Which of the following is correct for a first order reaction:

- a) $t_{0.5} = \frac{1.385}{K}$
c) $t_{0.75} = \frac{1.385}{K}$

- b) $t_{0.75} = 1.385 \times K$
d) $t_{0.5} = 1.385 \times K$

67. The activation energy of a reaction is:

- a) independent of temperature
c) decreases with temperature

- b) increase with temperature
d) none of the above

68. The degree of dissociation of 0.5% NaCl is 0.95 its molar mass is:

- a) 12 b) 23 c) 28 d) 30

69. Which of the following will have lowest freezing point:

- a) 0.1 molal solution of urea
b) 0.1 molal solution of sucrose
c) 0.1 molal solution of acetic acid
d) 0.1 molal solution of calcium chloride

70. The law of thermodynamics apply only to:

- a) matter in bulk
c) individual molecule

- b) individual atom
d) individual proton

71. Which of the following causes decrease in entropy:

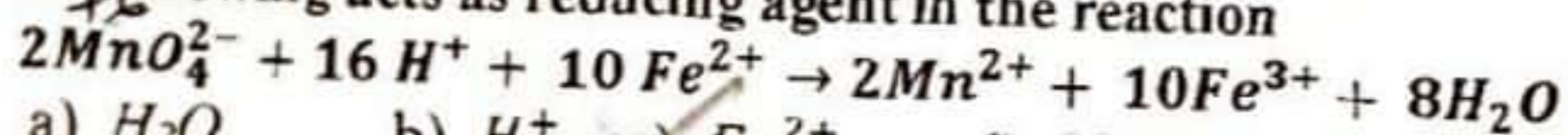
- a) crystallization of sucrose from solution
b) rusting of iron
c) conversion of ice to water
d) vaporization of camphor

72. If $E^0 Ag^+, Ag = 0.80 V$ $E^0 Cu^{2+}, Cu = 0.34 V$

$E^0 Zn^{2+}, Zn = 0.76 V$ the emf of the cell $Cu / Cu^{2+} // Ag^+ / Ag$ is:

- a) 0.56 V b) -0.48 V c) 0.46 V d) -0.43 V

73. Which of the following acts as reducing agent in the reaction



- a) H_2O b) H^+ c) Fe^{2+} d) MnO_2^-

74. The arrangement of sulphur in zinc blende and wurtzite structures respectively are:

- a) Hexagonal close packing and cubic close packing
b) Cubic close packing and hexagonal close packing
c) Simple cubic packing in both structures
d) Hexagonal close packing in both structures

75. An important method of removing soluble impurities of solution is:
a) Dialysis b) Peptization c) Sedimentation d) all the above

76. The temperature at which gas must be cooled before it can be liquefied by compression is termed as:

- a) Critical temperature b) Absolute temperature
c) Boyles temperature d) Zero temperature

77. According to VSEPR theory the shape of $[XeOF_5]^-$ is:

- a) Octahedral b) Trigonal bipyramidal
c) Square pyramidal d) Pentagonal monopyrimalal

78. The bond order for N_2 , O_2 , N_2^- and O_2^- varies as:

- a) $N_2 > N_2^- > O_2 > O_2^-$ b) $N_2 > O_2 > N_2^- > O_2^-$
c) $O_2 > N_2 > O_2^- > N_2^-$ d) $N_2^- > N_2 > O_2^- > O_2$

79. Penetration power of electron in an atom decreases from:

- a) nf to ns b) np to nf c) ns to nf d) ns to np

80. The maximum reducing power of Li atom in contact water is due to its maximum:

- a) standard oxidation potential b) ionization energy
c) sublimation energy d) hydration energy

81. Which of the following has lowest bond energy:

- a) F_2 b) N_2 c) O_2 d) H_2

82. The acid strength of halogen acids is:

- a) $HI > HBr > HF > HCl$ b) $HF > HCl > HBr > HI$
c) $HF < HCl > HBr > HI$ d) $HF < HCl < HBr < HI$

83. The electronic configuration of Gd^{3+} (at NO of Gd is 64) is:

- a) $[Xe]4f^05d^06s^0$ b) $[Xe]4f^75d^16s^2$
c) $[Xe]4f^75d^06s^0$ d) $[Xe]4f^05d^26s^2$

84. The complex with spin only magnetic moment of $-4.9 BM$ is:

- a) $[Fe(H_2O)_6]^{2+}$ b) $[Fe(CN)_6]^{3-}$
c) $[Fe(CN)_6]^{4-}$ d) $[Fe(H_2O)_6]^{3+}$

85. EAN of Fe^{2+} ion in $[Fe(CN)_6]^{4-}$ is equal to:

- a) 26 b) 36 c) 18 d) 54

86. The number of unpaired electrons in the square planer $[Pt(CN)_6]^{2-}$ are:

- a) 2 b) 1 c) 0 d) -3

87. The magnitude of Δ_0 in which of the following complexes of cobalt will be highest:

- a) $[Co(NH_3)_6]^{3+}$ b) $[Co(CN)_6]^{3-}$
c) $[Co(C_2O_4)_3]^{3-}$ d) $[Co(H_2O)_6]^{3+}$

88. Which of the following is ore of copper:

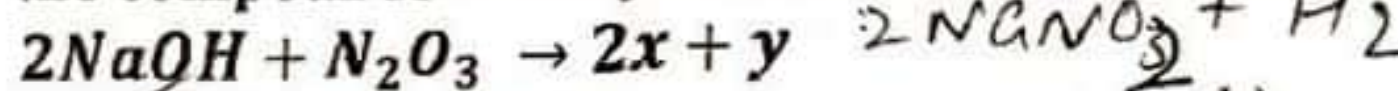
- a) Argentite b) Halmatite c) Malachite d) calcite

89. Blue vitrol when heated with 373K the product formed is:

- a) $\text{CuSO}_4 \cdot 3\text{H}_2\text{O}$
 c) $\text{CuSO}_4 \cdot \text{H}_2\text{O}$

- b) $\text{CuSO}_4 + \text{SO}_3$
 d) CuSO_4

90. Identify the compounds x and y in the following reaction



- a) $2\text{NaNO}_2 + \text{H}_2\text{O}$
 c) $\text{Na} + \text{NO}_2$

- b) $\text{NaN}_2 + \text{H}_2\text{O}$
 d) $\text{NaNO}_3 + \text{H}_2$

Biology (91-135)

91. Elephantiasis is caused by:

- a) ~~Wanchereria bancrofti~~
 c) *Plasmodium vivan*

- b) *Taenia solium*
 d) *Enterobius vermicularis*

92. The joint between atlas and axis is called:

- a) Pivot joint b) Angular joint c) Hingle joint d) Saddle joint

93. The stages between larval moults in an insect are called

- a) Pupae b) instars c) Grubs d) Caterpillers

94. Maximum diversity is found in

- a) Protozoa b) Arthropoda c) annelida d) Chordata

95. Sound producing organ of birds is

- a) Trachea b) larynx c) syrinx d) air sac

96. How many cleavage divisions are frequired for the formation of morula (32 celled)

- a) 3 b) 4 c) 5 d) 6

97. Which is the smallest known cell organelle

- a) Golgi complex b) Ribosome c) Lysosome d) Mitochondria

98. Fibrous lamina is a component of

- a) Nuclear membrane b) Nucleoplasm c) Nucleolus d) Chromatin

99. Fluid mosaic model was proposed by

- a) Danielle b) Robertson c) Singer & Nicholson d) Davson

100. Brain develops from

- a) Ectoderm b) ~~mesoderm~~ c) endoderm d) ectomesoderm

101. Desert snakes are

- a) Ammoniotelic b) ureotelic c) uricotelic d) aminotelic

102. Blood calcium level is maintained by

- a) Thyroxine b) PTH c) FSH d) LH

103. Study of fossils is called

- a) Ecology b) Ethology c) Palaentology d) Entmology

104. The unit of survival of the fittest is
a) Species b) Individual c) population d) Community
105. BCG vaccine is recommended for
a) TB b) Typhoid c) Cholera d) Polio
106. Who proposed classification of plants based on sex?
a) Bentham and Hooker b) Takhtajan c) Linnaeus d) Hutchinson
107. Which of the following is the correct sequence of Taxa,
a) Class, order, genus, family b) class, order, family, genus
c) Family, order, class, genus d) order, class, family, genus
108. Which of the following statement is correct in regard to herbarium
a) Collection of dry herbs b) collection of twigs of shrubs
c) Collection of twigs of trees d) collection of dry plants
109. The idea that O_2 comes from water not from CO_2 was first given by
a) Ruben and Kamen b) Van Niel
c) Arnon d) Hill
110. The type of taxonomic key which is most widely used in flora and manuals is
a) yoked key b) parallel key
c) bracket key d) none of the above
111. Fusion of protoplasts between two motile or non motile gametes is called
a) plasmolysis b) plasmogamy
c) oogamy d) karyogamy
112. Which of the following family is usually characterised by persistent sepals
a) Brassicaceae b) Labiatae
c) Solanaceae d) Rubiaceae
113. Which of the following is called imperfect fungi,
a) Deuteromycetes b) Basidiomycetes
c) Ascomycetes d) Phycomycetes
114. In *Cyperaceae*, each microspore mother cell forms,
a) only a single microspore b) two microspore
c) four microspores d) numerous microspores
115. Genes regulating many phenotypic characters are called
a) Complimentary genes b) Polygenetic genes
c) Pleiotropic genes d) Multiple genes
116. Chemical used to check the growth of algae in reservoirs is
a) alum b) brine
c) copper sulphate d) bleaching powder
117. In non endospermic seed, food is stored in,
a) chalazosperm b) cotyledons
c) perisperm d) embryo sac

118. Which of the following nitrogen base is not present in RNA,

- ☒ a) thiamine
- ☐ b) uracil
- ☐ c) guanine
- ☐ d) cytosine

119. Murashige and Skoog medium is used for

- ☒ a) raising plants through micropropagation
- ☐ b) culture of bacteria
- ☐ c) culture of protein rich cyanobacteria
- ☐ d) isolation of fungal strains

120. The nucleoprotein structures that occurs at the end of the chromosomes are

- ☐ a) centromeres
- ☒ b) telomeres
- ☐ c) centrosomes
- ☐ d) satellites

121. Increased heavy metal concentration in the environment is mainly

- ☐ a) natural
- ☒ b) due to activities of microorganisms
- ☐ c) anthropogenic
- ☐ d) none of the above

122. When a red female *Drosophila* is crossed with a white eyed male, the phenotype of progeny in F1 generation will be

- ☐ a) 1 red, 2 pink and 2 white eyed
- ☒ b) 3 red and 1 white eyed
- ☐ c) all red eyed
- ☐ d) all white eyed

123. Operon model for regulation of transcription was proposed by

- ☐ a) Meselson and Stahl in 1948
- ☒ b) Jacob and Monod in 1961
- ☐ c) Watson and Crick in 1951
- ☐ d) none of the above

124. Microorganisms employed to increase nutrient availability to crops are called

- ☐ a) chemical fertilizers
- ☒ b) biofertilizers
- ☐ c) synthetic fertilizers
- ☐ d) none of the above

125. Acid precipitation in the form of rain or snow is mainly caused by

- ☐ a) oxides of nitrogen
- ☐ b) carbon mono oxide
- ☐ c) chlorine and fluorine compounds
- ☒ d) sulphur dioxide and nitrogen dioxide

126. The flower may be described as perigynous, if the floral parts are

- ☐ a) inserted below the ovary
- ☒ b) inserted above the ovary
- ☐ c) surround the ovary
- ☐ d) none of the above

127. B.O.D stands for

- ☐ a) bacteria of damage
- ☐ b) before oxygen damage
- ☒ c) best organised drainage
- ☐ d) biological oxygen demand

128. The largest gene in the human genome is:

- ☐ a) TDF gene
- ☐ b) Insulin gene
- ☐ c) Haemoglobin
- ☒ d) Dystrophin gene

129. The metazoans with pseudocoelom are:

- ☐ a) Nematelminthes
- ☐ b) Annelids
- ☐ c) Molluscs
- ☒ d) Cnidarians

130. An autoimmune disease is:

- ☐ a) Haemophilia
- ☒ b) Myasthenia gravis
- ☐ c) AIDS
- ☐ d) Myxoedema

131. Which of the following enzymes is used in polymerase chain reaction (PCR)?
~~a) RNA polymerase~~ ~~b) Taq polymerase~~
~~c) Helicase~~ d) Reverse transcriptase

132. Industrial melanism is exhibited by:
 a) Dead leaf butterfly b) Stick insect ~~c) Peppered moth~~ d) Fruit fly

133. When does synapsis occur in meiosis?
~~a) Zygotene~~ b) Ceptotene c) ~~Diplotene~~ d) Pachytene

134. The HCl in gastric juice is secreted by:
~~a) Oxyntic cells~~ b) Panith cells c) Peptic cells d) Goblet cells

135. The life saving hormone is:
~~a) Aldosterone~~ b) Insulin c) Testosterone d) Growth hormone

Math (136-180)

136. $\lim_{x \rightarrow 0} \frac{(1 - \cos 2x) \sin 5x}{x^2 \sin 3x} = ?$
 a) $\frac{3}{10}$ b) $\frac{5}{6}$ c) $\frac{10}{3}$ d) $\frac{5}{3}$

137. $\lim_{x \rightarrow \infty} \frac{(2x-3x)(3x-4)}{(4x-5)(5x-6)} = ?$
 a) 6 b) $\frac{3}{5}$ c) $\frac{1}{10}$ d) $\frac{3}{10}$

138. If $y = \sin^{-1}(\cos x)$, then $\frac{dy}{dx} = ?$
 a) $-\cos^{-1}(\cos x) \sin x$ b) $\frac{1}{\sin x}$ c) -1 d) 1

139. The variance of the data 3, 5, 7, 9, ..., 99 is λ . Each observation of the data is increased by 9. Then the variance of the new data is
~~a) λ~~ b) $\lambda+9$ c) $\lambda-9$ d) 9λ

140. The standard deviation of 1, 2, 3, ..., n is
 a) $\sqrt{\frac{n^2+1}{12}}$ b) $\sqrt{\frac{n^2-1}{12}}$ c) $\sqrt{\frac{n^2-1}{6}}$ d) $\frac{n^2-1}{12}$

141. The maximum value of $p = x + 3y$ such that $2x + y \leq 20$, $x + 2y \leq 20$ and $x \geq 0$, $y \geq 0$ is
 a) 10 b) 60 c) 30 d) 40

142. The value of the determinant $\begin{vmatrix} y+z & x & 1 \\ z+x & y & 1 \\ x+y & z & 1 \end{vmatrix}$ is?
 a) $x+y+z$ b) xyz c) $xy+yz+zx$ d) 0

143. The determinant of a square matrix A of order 3 is 2. Then determinant of matrix 5A is
 a) 125 b) 250 c) 30 d) 10

144. Maximum value of $\frac{\log x}{x}$ in the interval $[2, \infty]$ is

- a) e ☒ b) $\frac{1}{e}$ c) $e+1$ d) $2e$

145. The slope of the normal to the curve $y = x^2 + 3x + 2$ at the point $(-2, 0)$ is

- ☒ a) 1 b) -1 c) $\frac{1}{3}$ d) $-\frac{1}{3}$

146. $\int e^x \left(\frac{1+\sin x}{1+\cos x} \right) dx =$

- ☒ a) $e^x \tan \frac{x}{2} + c$ b) $e^x \tan x + c$ c) $e^x \sec x + c$ d) $e^x \cos x + c$

147. $\frac{n}{c_1} + \frac{n}{c_2} + \frac{n}{c_3} + \dots + \frac{n}{c_n}$ where $n \in \mathbb{N} =$

- a) $\frac{n}{2}$ b) $\frac{n-1}{2}$ ☒ c) $\frac{n}{2} - 1$ d) None of these.

148. In an A.P., p^{th} term is q and q^{th} term is p then $(p+q)^{\text{th}}$ term is

- a) $p+q$ b) $p-q$ c) pq ☒ d) zero

149. For the system of equations $a_1x + b_1y + c_1z = 0$, $a_2x + b_2y + c_2z = 0$, $a_3x + b_3y + c_3z = 0$

If $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$, then this system has :

- ☒ a) Infinity many solutions b) only trivial solutions
c) no solutions d) one trivial and one non-trivial solutions.

150. The probability that a leap year may contain 53 Fridays is

- ☒ a) $\frac{1}{7}$ ☒ b) $\frac{2}{7}$ c) $\frac{3}{7}$ d) $\frac{1}{12}$

151. For non-singular matrices A, B, C of same order, $(A B^{-1} C^{-1})^{-1} ?$

- a) $A^{-1} B^{-1} C$ b) $C^{-1} B^{-1} A^{-1}$ ☒ c) $C B A^{-1}$ d) $C^{-1} B A^{-1}$

152. The adjoint of matrix $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ is

- ☒ a) $\begin{bmatrix} \cos \theta & -\sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ b) $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$
c) $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ d) $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

153. The modulus function $|x|$ is

- a) Continuous and differentiable for all values of x
b) neither continuous nor differentiable for all values of x
c) is differentiable at $x=0$
☒ d) is not differentiable at $x=0$

154. If $f(x) = \sqrt{\frac{x-\sin x}{x+\cos 2x}}$, then $\lim_{n \rightarrow \infty} f(x)$ is

- a) 0 ☒ b) 1 c) ∞ d) does not exist

155. The value of $\int_0^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx =$
 a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) π d) $\frac{\pi}{6}$
156. The value of integral $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx =$
 a) $2\sqrt{\sec x} + c$ b) $\frac{2}{\sqrt{\tan x}} + c$ c) $\sqrt{\tan x} + c$ d) $2\sqrt{\tan x} + c$
157. Integration factor of differential equation $\frac{dy}{dx} + \cot x \cdot y = \operatorname{cosec} x$ is
 a) $\log \sin x$ b) $-\operatorname{cosec}^2 x$ c) $\sin x$ d) $\operatorname{cosec} x$
158. A box contains 20 cards bearing numbers from 1 to 20. A card is drawn at random from the box. The probability that the card drawn is an odd prime number is?
 a) $\frac{7}{20}$ b) $\frac{8}{20}$ c) $\frac{9}{20}$ d) None of these
159. In $\cos x + \cos y + \cos z = 3$ then $(\sin x + \cos x + \cos y) \cos y + \sin z =$
 a) 1 b) 2 c) 4 d) 3
160. How many 8-letter words (Letter distinct) can be formed from the letters of the word "DOMESTIC" so that all vowels occur together?
 a) 4140 b) 4320 c) 432 d) 43
161. There are 7 points on the plane surface. Out of these 4 points are collinear. How many straight lines can be formed by joining these points?
 a) 21 b) 16 c) 14 d) none of these.
162. The lines represented by $2x - 3y + 5 = 0$ and $3x + 2y + 7 = 0$ are
 a) parallel b) perpendicular c) identical d) passing through origin
163. Angle between the lines $\frac{x}{a} + \frac{y}{b} = 1$ and $\frac{x}{a} - \frac{y}{b} = 1$ is
 a) $2 \tan^{-1} \frac{b}{a}$ b) $\tan^{-1} \frac{2ab}{a^2 + b^2}$ c) $\tan^{-1} \frac{a^2 - b^2}{a^2 + b^2}$ d) None of these
164. The radius of the circle $x^2 + y^2 + 4x - 8y - 29 = 0$ is
 a) 7 b) 5 c) 3 d) 4
165. The directrix of parabola $y^2 = 16x$ is
 a) $x = 4$ b) $y = 4$ c) $y = -4$ d) $x = -4$
166. If e is eccentricity of a hyperbola then
 a) $e = 1$ b) $e < 1$ c) $e > 1$ d) $e = 0$
167. The maximum value of $P = 6x + 8y$ such that $2x + y \leq 30$, $x + 2y \leq 24$, $x \geq 0$, $y \geq 0$ is
 a) 90 b) 120 c) 96 d) none of these
168. $y = x + x^2 + x^3 + \dots$ to ∞ , then $x = ?$
 a) $\frac{y}{1+y}$ b) $\frac{1-y}{y}$ c) $\frac{y}{1-y}$ d) None of these

169. The third term of G.P is 4. Then product of the first five term is
 a) 1024 ☒ b) 243 c) 3125 d) 132
170. The angle between two lines $\frac{x+1}{2} = \frac{y+3}{2} = \frac{z-y}{-1}$ and $\frac{x-4}{1} = \frac{y+4}{2} = \frac{z+1}{2}$ is
 a) $\cos^{-1} \frac{1}{9}$ b) $\cos^{-1} \frac{2}{9}$ c) $\cos^{-1} \frac{1}{3}$ ☒ d) $\cos^{-1} \frac{4}{9}$
171. In a class of 100 students, 55 students have passed in math and 67 passed in physics then number of students who passed in physics only is
 a) 22 b) 33 ☒ c) 45 d) 10
172. R is a relation on set of natural numbers,
 $R = \{(m, n) : 2m + 3n = 30; m, n \in N\}$ then domain of R is
 a) $\{2, 4, 6, 8\}$ ☒ b) $\{3, 6, 9, 12\}$ c) $\{3, 7, 11, 15\}$ d) $\{3, 6, 9, 12, 15\}$
173. The conjugate of $\frac{(2+i)^2}{3+i}$ in the form $a + ib$ is
☒ a) $\frac{13}{10} + i(\frac{-15}{2})$ b) $\frac{13}{2} + i(\frac{-15}{2})$ c) $\frac{13}{10} + \frac{i+9}{10}$ ☒ d) $\frac{13}{10} + (\frac{i9}{10})$
174. Middle term in the expansion of $(1+3x+3x^2+x^3)^6$ is
 a) 4th term ☒ b) 10th term c) 3rd term d) None of these
175. $\int_0^{\frac{\pi}{4}} \frac{e^{\tan x}}{\cos^2 x} dx = ?$
☒ a) $e-1$ b) $e^{-1}-1$ c) e^2-1 d) $e^{-2}-1$
176. $\vec{a} = 2\hat{i} - 3\hat{j} - \hat{k}$, $\vec{b} = \hat{i} + 4\hat{j} - 2\hat{k}$ then $\vec{a} \times \vec{b} = ?$
 a) $10\hat{i} - 2\hat{j} + 11\hat{k}$ b) $10\hat{i} - 3\hat{j} - 11\hat{k}$
☒ c) $10\hat{i} + 3\hat{j} + 11\hat{k}$ d) $10\hat{i} - 3\hat{j} - \hat{k}$
177. $\hat{a}, \hat{b}, \hat{c}$ are mutually perpendicular unit vectors, then $|\hat{a} + \hat{b} + \hat{c}| = ?$
☒ a) $\sqrt{3}$ b) 3 c) 0 d) 1
178. If a straight line makes angle α, β, γ with x-axis, y-axis, and z-axis, respectively, then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma =$
 a) 1 b) 3 ☒ c) 2 d) 4
179. If $\cos \theta = \cos \alpha$, then general value of θ is
 a) $\theta = n\pi + (-1)^n \alpha$ b) $\theta = n\pi + \alpha$
 c) $\theta = 2n\pi + \alpha$ ☒ d) $\theta = 2n\pi \pm \alpha$
 (n being an integer)
180. If $y = x^x$ then $\frac{dy}{dx} = ?$
 a) $x \cdot x^{x-1}$ b) $x^x \cdot \log_e x$ c) $x^x(1 - \log_e x)$ ☒ d) $x^x(1 + \log_e x)$

$$\log y = \log x^x$$

$$\frac{1}{y} = x \cdot \log_e x$$